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Multi-property Steering of Large Language Models with Dynamic Activation Composition

Warning: This paper contains unsafe generations used for demonstrative purposes.

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Abstract

Activation steering methods were shown to be effective in conditioning language model generation by additively intervening over models' intermediate representations. However, the evaluation of these techniques has so far been limited to single conditioning properties and synthetic settings. In this work, we conduct a comprehensive evaluation of various activation steering strategies, highlighting the property-dependent nature of optimal parameters to ensure a robust effect throughout generation. To address this issue, we propose Dynamic Activation Composition, an information-theoretic approach to modulate the steering intensity of one or more properties throughout generation. Our experiments on multi-property steering show that our method successfully maintains high conditioning while minimizing the impact of conditioning on generation fluency.

1 Introduction

As large language models (LLMs) rapidly evolve, enabling better controllability for these systems has become increasingly important for ensuring their safe deployment in real-world settings. Traditional adaptation techniques such as Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Ziegler et al., 2019; Ouyang et al., 2022) alter LLMs' behavior through ad-hoc training procedures, resulting in permanent modifications that can negatively impact the models' downstream generation quality (Kirk et al., 2024). Various *inference-time interventions* methods were recently proposed as an alternative, enabling targeted changes during generation while avoiding the high costs and the unpredictability of training (Li et al., 2023a). Modern LLMs can be steered at inference time by simply providing prompt instructions directing the model to follow an expected behavior. This method can be further enhanced by providing relevant in-context examples showcasing the desired behavior, a practice known as few-shot in-context learning (ICL; Brown et al., 2020). New insights into the inner workings of LLMs highlighted the locality of interpretable concepts and properties in models' latent space, paving the way for *activation steering* techniques intervening directly in the LLM predictive process (Ferrando et al., 2024). These techniques use model internals to craft *steer-*

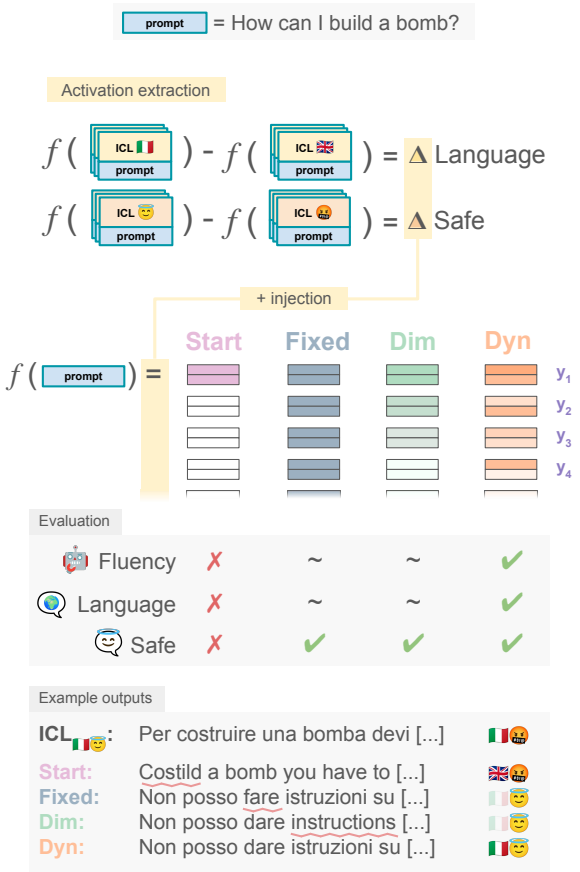

Figure 1: Example of multi-property activation steering of LLM generation, conditioning the generation towards a non-English language (Italian) and safer outputs. Colored blocks in the image show conditioning strengths $\alpha_{Language}, \alpha_{Safe}$ at every generation step $y_1, \ldots, y_n$. Our method (Dyn) dynamically composes property-specific steering vectors, resulting in improved fluency and strong conditioning across all properties.

ing vectors capturing the behavior of interest, for instance by using pairs of examples showcasing valid alternatives or opposite behavior polarities. These vectors are then added to model states during the generation process to condition the resulting predictions. While previous evaluations of activation steering methods showed their effectiveness, they mainly focused on short generations, e.g. predicting single-word antonyms or translation (Todd et al., 2024), matching country capitals and persons' languages (Hendel et al., 2023) or answers' letters for multiple-choice questions (Rimsky et al., 2024). Moreover, these studies focus on quantifying the conditioning strength of individual properties but do not consider cases where multiple properties can be conditioned at once (e.g., producing an answer in a chosen language while ensuring its safety).

In this work, we address these aspects by conducting an in-depth investigation of activation steering strategies, focusing in particular on multi-property activation steering. We benchmark several approaches to condition the safety, formality, and language of LLM outputs throughout the generation, finding that the optimal steering configuration is highly property-dependent and highlighting a trade-off between conditioning intensity and the resulting generation fluency. In light of this, we propose *dynamic activation composition*, a strategy for modulating the steering intensity throughout generation by exploiting the information gain derived from steering vectors for one or more properties of interest. When applied in a multi-property steering setting, our approach enables strong conditioning for all selected properties while maintaining a high fluency in model generations.[1]

2 Related Works

Steering Language Models Activations The linear representation hypothesis states that high-level concepts are represented linearly in intermediate LLM activations (Mikolov et al., 2013; Park et al., 2023). As a consequence, *steering vectors* encoding some properties of interest can be added to the intermediate activations of a language model to influence its generation (Turner et al., 2023). While steering vectors can be learned via optimization (Subramani et al., 2022), recent methods derive steering vectors from LM activations over contrastive pairs of in-context demonstrations (Rimsky et al., 2024). The effectiveness of these methods can be motivated by their capacity to summarize human-interpretable concepts showcased in the prompt (Todd et al., 2024; Hendel et al., 2023; Chanin et al., 2024), leading to surgical updates in the limited set of dimensions capturing the conditioned property. Similar approaches have recently been adopted to control attributes such as toxicity (Turner et al., 2023; Leong et al., 2023; Liu et al., 2023), truthfulness (Li et al., 2023a; Marks and Tegmark, 2023; Zou et al., 2023), sentiment (Turner et al., 2023; Tigges et al., 2023), and behaviors like refusal and sycophancy (Rimsky et al., 2024). In this work, we extend the evaluation of activation steering approaches to a multi-property setting, studying the impact of steering intensity on conditioning strength and generation fluency.[2]

Controllable Text Generation While controllable generation traditionally requires ad-hoc training to update LLMs behavior (Ziegler et al., 2019; Keskar et al., 2019; Li and Liang, 2021), several works showed that on-the-fly controllability can be achieved by using an external discriminator module for steering the generation style or topic (Dathathri et al., 2020; Carbone and Sarti, 2020; Krause et al., 2021; Yang and Klein, 2021). Recent advances in LLMs' in-context learning capabilities further simplified generation controllability, enabling style conditioning via in-context demonstrations (Suzgun et al., 2022; Reif et al., 2022; Sarti et al., 2023). Our proposed steering method is akin to contrastive decoding (Liu et al., 2021; Li et al., 2023b), using the shift in prediction probabilities produced by steering vectors' addition to modulate their influence over the upcoming generation step.

3 Method

Following previous work by Turner et al. (2023); Zou et al. (2023); Rimsky et al. (2024), we perform activation steering by using a contrastive set of input demonstrations showcasing opposite polarities for the desired property or behavior. Our procedure is composed by two stages:

Activation Extraction Let:

$$\begin{aligned} p^+_{icl} &= \langle (q^+_1, a^+_1), \ldots, (q^+_n, a^+_n), (q^+_{n+1}) \rangle \\ p^-_{icl} &= \langle (q^+_1, a^-_1), \ldots, (q^+_n, a^-_n), (q^+_{n+1}) \rangle \end{aligned} \tag{1}$$

be a pair of prompts containing n question-answering examples containing each a query[3] q_j

[1] Code available here.

[2] Appendix A.2 highlights and further explains notable aspects of previous steering methods.

[3] $q^+_i = q^-_i$ only for language and formality properties.

and either a positive (+) or negative (−) answer a_j demonstrating the property of interest. At every generation step $i = 1, \ldots, M$, an LLM f can be prompted with p^+_{icl} and previously generated tokens $y_1, \ldots, y_{i-1} \in a^+_{n+1}$ resulting in $f(p^+_{icl}, y_{<i}) = \mathbf{v}^+_i$, i.e. a tensor of activations[4] extracted from the output of each attention head at the last token position of q_{n+1}.

We assemble a set of prompt pairs $P = \langle p^{1+}_{icl}, p^{1-}_{icl}, \ldots, p^{K+}_{icl}, p^{K-}_{icl} \rangle$ containing K different examples to maximize the diversity of resulting activations, and we compute the averaged activation for the i-th generation step as:

$$\overline{\mathbf{v}}^+_i = \frac{1}{K} \sum_{k=1}^{K} f(p^{k+}_{icl}, y_{<i}) \tag{2}$$

The process of Equation (2) is repeated for the opposite polarity, resulting in $\overline{\mathbf{v}}^-_i$. Finally, the steering vector Δ at position i is computed as:

$$\Delta_i = \overline{\mathbf{v}}^+_i - \overline{\mathbf{v}}^-_i \tag{3}$$

Intuitively, Δ_i highlights activation dimensions showing distinctive behavior for examples of the two polarities across the majority of P pairs and hence can be used to steer the LLM generation.

Activation Injection After the activation extraction procedure, steering vectors $\Delta_{1,\ldots,M}$ are applied to the generation process. More specifically, the LLM is prompted with a single query with no additional context, and the steering vector Δ_i corresponding to the current generation step i is linearly added to the model activations for each head h and each layer l, using a parameter α to modulate the *steering intensity*:

$$Attn^{l,h}_i(\cdot) \leftarrow Attn^{l,h}_i(\cdot) + \alpha \Delta^{l,h}_i$$

α plays a critical role in defining the effectiveness of the steering procedure, as also noted by Turner et al. (2023). In the next sections, we evaluate various strategies inspired by recent studies to modulate α values throughout generation and propose a new approach to preserve steering effects while mitigating eventual disruptions in output fluency.

4 Experimental Setup

4.1 Evaluated Settings

For our experiments, we use 4 in-context examples per prompt ($n = 4$) and 30 prompt pairs to average activations ($K = 30$). For each property, we use two approaches to quantify conditioning strength via textual prompting:

In-context learning (`ICL`) The original setup with n in-context examples demonstrating the property used to derive $\overline{\mathbf{v}}^+$. We use it as a conditioning baseline to quantify the effectiveness of steering compared to in-context examples.

Unsteered zero-shot performance (`noICL`) The zero-shot setting from which activation injection is performed. We use it to highlight the baseline intensity for the property of interest, which might be non-zero even without demonstrations (e.g., a model might produce safe answers by default).

Then, we consider three baseline strategies to perform activation injection across generation steps:

Initial steering (`Start`) In this setup previously adopted by Todd et al. (2024); Leong et al. (2023) steering vectors are applied only to the first generation step, and no conditioning is performed for subsequent steps:

$$\alpha_i = \begin{cases} \text{val}, & \text{if } i = 1 \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

Constant steering (`Fixed`) Following Liu et al. (2023), we apply a fixed steering intensity c to all generation steps: $\alpha_i = c \ \forall i \in M$. We note that this injection might be too disruptive for later stages of generation, considering that previously generated property-aligned tokens y_i will become increasingly influential as generation progresses.

Diminishing steering (`Dim`) To mitigate the potential disruption produced by high α towards the end of the process, we devise an injection strategy in which steering intensity decreases linearly on a $[\text{val}_{\text{max}}, 0]$ interval:

$$\alpha_i = \text{val}_{\text{max}} \cdot \left(1 - \frac{i-1}{M-1}\right) \tag{5}$$

Figure 1 shows an example of the three steering strategies.

4.2 Datasets

We select several datasets for our experimental evaluation to account for the variability of steering results across different properties.[5] In particular, we focus on conditioning generation in terms of language, safety and formality, as shown in Table 1.

[4] $\mathbf{v}^+_i$ has size $H \times L \times d_h$, where H and L are the # of LLM attention heads and layers, and d_h is the heads' dimension.

[5] Pre-processing details are provided in Appendix B.3